

Multirotor Recovery Dynamics

A study and defence reference

Understanding every decision in the project: the research question, the physics, the statistics, the evidence discipline, and the corrections that shaped it.

Repository: [500ft/multirotor-recovery-dynamics](#)
Compiled 27 September 2026

How to use this document

This is not a summary. It is the reasoning behind the project, written so that you can reconstruct any decision under questioning rather than recall a conclusion.

Each part ends with a **defence question**. Those are the real test. If you can answer them without looking, you understand the part; if you cannot, reading it again will not help as much as working the exercise it points to.

One habit matters more than any fact in here. When challenged on a number, answer in four steps, in this order: **(1)** what kind of number is it - requirement, calculation, assumption or measurement? **(2)** where does it come from? **(3)** what happens if it is wrong? **(4)** how would you close it? If step 4 has no answer, the number is a liability - and saying so before your examiner does is the strongest move available to you.

Every figure quoted here was read from the repository at compile time, not recalled. Where a number is assumed rather than measured, this document says so, because that distinction is the project's central discipline.

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Part 1 - The research question

The project does **not** ask whether this aircraft can recover after a rotor fails. It asks a sharper question:

After a partial loss of propulsion authority, which recovery action maximises the probability of a survivable landing from a given post-failure state, and does the dedicated recovery mechanism enlarge the survivable set beyond what thrust reallocation alone achieves?

Three consequences follow from that shape, and each one drove a design decision.

It is a comparison

A comparison needs at least two actions and a criterion capable of separating them. That is why there are three modelled actions (reallocation, the guard-enabled mechanism, a descent device) and why a great deal of care goes into the landing criterion. A study that only demonstrates one controller cannot answer it.

It is conditional on a state

The answer is a function of where you are when the failure happens: height, vertical speed, tumble rate, and how long detection takes. A result without its state is meaningless. This is why the study sweeps a grid of post-failure states rather than reporting a single success rate.

It admits a negative result

The kill criterion was registered **before** any data existed: the mechanism is justified only if its exact lower bound exceeds reallocation's exact upper bound somewhere in the primary cells. Because that rule was frozen in advance, 'the mechanism is not justified' is a publishable finding rather than a failure. This is the single most important structural decision in the project.

What the study actually delivers today

An important correction, applied after external critique: the comparison varies **installed hardware** between its 'actions' - the reallocation case removes the guard's mass and rim inertia, the mechanism case keeps it. An airborne vehicle cannot choose to have launched without its guard.

So what exists is a **comparison of design packages**, not a runtime policy. The ranking output tells you which aircraft to build, not which action to take in flight. A genuine runtime policy requires one fixed configuration and an action set that a single aircraft can actually execute.

Defence question. Why can this study not currently claim to produce a runtime policy, and what exactly would have to change for it to do so?

Part 2 - Evidence discipline

This is the repository's distinguishing feature and the first thing a reviewer will probe. Learn it before any technical part, because every number is read through it.

Two independent labels

Every consequential quantity carries a **provenance** and, separately, an **evidence status**. They are not the same axis, and collapsing them is the most common way engineering documents mislead.

Provenance	Meaning
requirement	An imposed target - regulatory, contractual or self-set
measured input	A physical measurement used as an input
sourced assumption	A value taken from literature or a datasheet
calculated result	Derived from other quantities
selected design value	A choice, informed by the above
measured result	A physical measurement of an outcome
provisional estimate	A placeholder, explicitly marked

Evidence status runs alongside: **unverified**, **analytically assessed**, or **physically tested**. A selected design value can be unverified. A calculated result is never a measured result.

No quantity in this repository has status 'measured'. There is no accepted physical measurement yet. The traceability index states this explicitly rather than leaving a reader to infer it. Adding documentation never moves a number to 'measured' - only recorded physical evidence does.

The three tiers found by audit

Tier	Meaning	Example
A	Derived, with the derivation recorded	0.020 N.m authority threshold
B	Purpose stated, derivation absent	T/W >= 2.0; the 225 g abort threshold
C	Bare constant, no requirement ID	5 g unmodelled-hardware allowance

Tier B is subtle and worth dwelling on. The requirements file carries a one-line rationale for every requirement - but those lines say what a number is *for*, not *why that* number. 'Minimum static control authority' does not tell you why 2.0 rather than 1.8. That gap is what the audit exists to surface.

Worked example: tracing one number end to end

The best-evidenced number in the project is the authority threshold. Trace it and you have seen the whole system:

```
225 g full reserve -> 4.41 N total thrust
at 75% collective -> 3.31 N
5 mm thrust-line offset -> 3.31 x 0.005 = 0.0166 N.m disturbance
threshold 0.020 N.m -> about 20% margin over that disturbance
```

Now ask the question that matters: **what would make this wrong?** It depends on a 5 mm offset assumption, a 225 g requirement, and a mixer model using an assumed 60 mm arm. Three assumptions, none measured. Every number in the project decomposes this way, and being able to do it reflexively is what defending the work consists of.

Defence question. A reviewer says 'your simulation shows 90% survival.' What is wrong with that sentence, in three separate ways?

Part 3 - Vehicle physics

The simulation is a genuine 6-DoF rigid-body model: quaternion attitude, translational dynamics with attitude-coupled thrust, and quadratic drag.

Why quaternions

Euler angles suffer gimbal lock - at 90 degrees of pitch, two axes align and the representation loses a degree of freedom. A recovery study deliberately drives the vehicle through large attitudes, so that singularity would sit in the middle of the region of interest. Quaternions have no such singularity, renormalise cheaply, and integrate as:

$$\dot{q} = 0.5 * q(x) [0, \omega] \quad \text{then renormalise } q$$

The cost is that a quaternion is not directly readable. The simulation therefore converts to a rotation matrix for the parts that need physical meaning - tilt is recovered as the angle between body-up and world-up.

Euler's equations and gyroscopic coupling

$$I * \dot{\omega} = \tau - \omega \times (I * \omega)$$

The second term is the one that matters after a rotor failure. A spinning body resists being re-pointed: angular momentum about one axis produces a torque about a perpendicular axis when you try to rotate it. As post-failure spin builds, this precession term grows and competes with the control torque.

The A2 controller variant tried to cancel exactly this term with feedforward. It did not help, for reasons covered in Part 5.

The mixer - where control authority comes from

Four rotors sit at (+/-a, +/-a) with per-rotor thrust bounded to $[0, T_{\max}/4]$. To roll without changing collective, raise one pair by d and lower the other by d . The available differential is limited by whichever bound you hit first:

$$d_{\max} = \min(T/4, T_{\max}/4 - T/4)$$
$$\tau_{\text{rp}}(T) = 2 * \sqrt{2} * \text{arm} * d_{\max}$$

Read that carefully, because it has a consequence people find surprising: **differential authority is zero at both zero and full collective**. At zero thrust you cannot lower a rotor; at full thrust you cannot raise one. Authority peaks exactly in between.

The derivation behind thrust-to-weight ≥ 2.0

Setting the two terms of $d_{\max}$ equal:

$$T/4 = T_{\max}/4 - T/4 \quad \Rightarrow \quad T = T_{\max} / 2$$

Hover requires $T = mg$. Placing hover at the authority optimum therefore requires:

$$m * g = T_{\max} / 2 \quad \Rightarrow \quad T_{\max} / (m * g) = 2.0$$

T/W = 2.0 places hover exactly at the collective of maximum roll/pitch authority. Verified numerically against the implementation: the peak sits at 0.5000 of $T_{\max}$ (0.08910 N.m at $T_{\max} = 4.2$ N, arm = 60 mm), i.e. $T/W = 2.0000$. This is a *retrospective assessment* - it shows the requirement is defensible, not that this was the original reason for choosing it.

Airmode

Because authority vanishes at full collective, the controller never commands it. The implementation caps collective below maximum so differential headroom cannot reach zero. This costs peak climb thrust and buys

controllability - the standard trade in multirotor flight control, and the reason an aggressive quadrotor stays controllable when a naive one flips.

Defence question. Why does thrust-to-weight of exactly 2.0 place hover at the point of maximum roll and pitch authority? Derive it.

Part 4 - Failure classes and control allocation

The wrench map

Allocation maps per-rotor thrusts to a body wrench:

$$\mathbf{w} = \mathbf{B} \mathbf{f}, \quad \mathbf{w} = (T, \tau_x, \tau_y, \tau_z), \quad 0 \leq f_i \leq \text{cap}$$

A failed rotor is a column forced to zero. Partial authority is a reduced cap on all four. These are different faults and the distinction matters - see Part 11.

Reduced-attitude allocation: sacrificing yaw

After rotor loss the drag-torque budget cannot balance yaw, so the vehicle spins about body z. Commanding yaw anyway would spend scarce authority on a hopeless objective. The allocator therefore commands only (T, τ_x, τ_y) .

Crucially, the **achieved** yaw torque from the resulting thrust pattern is still fed to the dynamics. The spin is simulated, not ignored. Dropping yaw from the command is a control decision; pretending the spin does not happen would be a modelling error.

Topology matters more than count

Two failure classes lose the same collective and behave completely differently, because of spin directions. Rotors alternate CCW/CW around the frame.

Class	Survivors	Spins	Yaw drag	Consequence
two_adjacent	0 and 1	opposite	cancels	No yaw spin, but no roll trim exists
two_opposite	0 and 2	same	adds	Persistent yaw spin; torque axis fixed

One rotor out

With one rotor dead, zero roll/pitch torque forces the opposite rotor to zero as well - flight continues on the remaining diagonal pair. The **balanced** collective ceiling is therefore $2 \times f_{\text{max}}$, not $3 \times f_{\text{max}}$: the odd rotor's thrust cannot be used without unbalancing the vehicle.

Two adjacent out

Both survivors sit on the same side. Any positive collective produces a positive roll torque, so an exact zero-roll trim does not exist at any positive thrust. That is a statement about *static trim*, not about flight - a dynamic spinning trajectory is a different question, and the repository is careful not to conflate them.

What the allocator actually is

The implementation performs a least-squares solve, pins saturated rotors at their bounds, and re-solves the free ones with a priority weighting. This is a named, thirty-year-old method: the **redistributed pseudo-inverse** (Virnig and Bodden, 1994), in the cascaded-generalized-inverse lineage.

The literature documents its quality honestly. Haerkegaard (2002) states that RPI-type methods 'only deliver approximate and sometimes unreliable solutions', while active-set solvers reach the true optimum at comparable runtime. The correct position is to name the method, cite it, justify it on implementation cost, and **not** present it as an invention.

A second honest caveat: the torque-priority weight of 10 mixes a force residual in newtons with a moment residual in newton-metres. No source prescribes a dimensional rule for that. Normalising it would be a contribution to document, not a citation to borrow.

Defence question. Why can a quadrotor with two adjacent rotors out not hold a zero-roll trim at any positive thrust - and why is that not the same as saying it cannot fly?

Part 5 - The controller and its boundary

A saturated PD attitude loop drives body-up toward world-up, with a separate vertical loop. In descent mode the vertical loop brakes while tilt is large and only tracks the touchdown rate once the vehicle is upright - descending while still righting spends the height the arrest needs.

The A2 variant

Variant A2 added two things at once: gyroscopic feedforward (cancelling the $\omega \times I \omega$ term in the tilt plane) and a yaw rotational-drag plant term so the post-failure spin reaches a bounded terminal rate.

Registered hypothesis, before the sweep: A2 would help most in tall cells, where spin has time to build, and be indistinguishable at low height. The result contradicted it. Across 192 paired comparisons the spin-aware variant won **zero** discordant pairs; the baseline won all 15 that existed.

That is a stronger result than 'no significant difference'. It is directional evidence that the feedforward is mildly harmful in the cells where the two differ at all - and because the hypothesis was registered in advance, the finding is interpretable rather than a post-hoc rationalisation.

The boundary finding

At a 6 rad/s tumble the rotor-out classes fail at every tested height. Tilt passes 78 degrees before the reduced torque authority can arrest the rotation, independently of the descent that follows.

This is a **control-family limit, not a tuning problem**. The registered next step is a spin-aware reduced-attitude controller - and that is *adoption of known prior art* (the primary-axis relaxed-hover solution, which is also patented), not an invention. Saying so is not modesty; claiming otherwise is the kind of error a reviewer will find.

Defence question. The A2 variant made outcomes slightly worse. Why is that a more useful result than 'no significant difference'?

Part 6 - Statistics

The area most likely to be attacked, and where this project has already made and corrected real errors. Understand the reasoning, not just the method names.

Why Clopper-Pearson

It is the *exact* binomial interval, obtained by inverting the equal-tailed binomial test. Its guarantee is coverage of at least $1 - \alpha$ for **every** true p .

Know the standard criticism, because a good examiner will raise it: Brown, Cai and DasGupta (2001) call it 'wastefully conservative' and recommend Wilson or Agresti-Coull for general use. The honest defence has three parts:

Argument	Why it applies here
Coverage must never fall below nominal	This is a safety-adjacent claim; an interval that under-covers at some p is unusable
The grid lands near $p = 0$ and $p = 1$ constantly	Wald degenerates there; many cells are exactly 0 or exactly 1
The direction of conservatism is favourable	A conservative lower bound on survival errs toward understating it

Sidedness - an error this project made

A 95% lower bound combined with a 95% upper bound on **one** proportion is a 90% central interval, because each tail carries 5%. That much is textbook.

But the kill criterion compares a lower bound on the *mechanism's* probability against an upper bound on *reallocation's* - **two different proportions**. That is not a central interval for their difference. It is not an interval at all: it is a decision rule. The repository originally described it as a 90% central construction, which was wrong, and the correction is recorded.

Pairing

Running both arms on common random numbers induces positive covariance. Since

$$\text{Var}(A - B) = \text{Var}(A) + \text{Var}(B) - 2 \cdot \text{Cov}(A, B)$$

that sharpens the comparison. Note carefully what this is and is not: it is **variance reduction**, not a repair. The earlier independent-draw design was valid, just low-powered. An earlier version of the repository claimed the unpaired analysis 'manufactured' its null result; that overstated the case and was withdrawn.

Analysing paired binary outcomes

Use McNemar's conditioning on the discordant pairs. Report **both**:

Quantity	Meaning	Trap
π	P(a discordant pair favours A)	Conditional - not the improvement in success
$\Delta = (n_{10} - n_{01})/N$	Unconditional difference in success	The quantity people actually mean

What the bounds do not cover

The intervals are **conditional Monte Carlo sampling bounds**. They quantify finite sampling under this model, these distributions and this landing proxy. They do not cover model error, scenario error, or real-world reliability - and more samples shrink only the first of those.

A related trap: per-cell 95% intervals over many cells are not 95% simultaneous. Any claim about the *set* needs either an explicit 'per-cell, no simultaneity claimed' or a multiplicity adjustment.

Defence question. Zero discordant pairs in 300 paired trials. What may you conclude, and what may you explicitly not conclude?

Part 7 - Study design and its failure modes

The registered criteria

Quantity	Value	Status
Bare landing criterion	2.0 m/s impact, 30 deg tilt	PREREGISTERED-ASSUMED
Guarded criterion	2.5 m/s, 60 deg	Assumed (owner input outstanding)
Mass rollup (best/nominal/worst)	123.86 / 135.66 / 156.0 g	Calculated from itemised budget
Frozen maximum	165.0 g	Calculated: roundup_to_5g(worst + 5 g)
Abort threshold	225 g	Selected round buffer, not allocated

The landing criterion deserves attention because a reviewer will push on it. It cannot rest on human-injury data: at 165 g and 3 m/s the vehicle carries 0.74 J, against a most-conservative published yardstick of 14.9 J - non-binding by roughly 27 times, and that yardstick is itself a rigid-object equivalence a frangible airframe clears by another order of magnitude.

Landing-gear certification gives useful *context* - 14 CFR 27.725 and 27.727 imply roughly 2.55 and 3.12 m/s - but those are qualification tests for other vehicle classes at other scales. They do not validate a limit for a sub-250 g plastic drone, and the criterion keeps its PREREGISTERED-ASSUMED label. Only a drop test of the actual vehicle changes that.

Saturation: the lesson worth internalising

A measure that cannot move tells you nothing. Of 96 class-cell combinations, **71 are saturated** at 0 or 1, and all four registered primary cells are pinned at zero.

The scenario diagnostic therefore returned 'no discordant pairs' everywhere - **by saturation, not by agreement**. A full rerun on those cells would have consumed 7200 trajectories and measured nothing. That only became visible because a 288-trajectory diagnostic was run first.

The scenario distinction

The simulator cuts **all four** motors during detection latency, then engages the controller. That is a release/startup transient - a vehicle dropped with motors off - not an in-flight failure where healthy motors keep running until detection.

Every rotor-out number in the project is conditional on that reading. The mode is now named in code and emitted in every results file, with a test that fails if the dynamics stop matching the label.

Three arms, to avoid a combined-change result

Arm	Before the switch	Recovery starts
L (legacy)	all rotors zero	detection + spool
H (matched)	failed zero, healthy hold trim	detection + spool
I (immediate)	failed zero, healthy hold trim	detection

H minus L isolates the actuator history; I minus H isolates transition timing; I minus L is the combined effect. Splitting them prevents reporting one number that mixes two changes - a mistake the project had already made once with A2.

Defence question. Why is choosing new cells by looking at existing results legitimate for designing an experiment, but not a confirmatory test of the same hypothesis?

Part 8 - Measurement and uncertainty

Calibration is not adjustment

The international vocabulary of metrology (VIM 2.39, Note 2) is explicit: calibration must not be confused with *adjustment* of a measuring system, 'often mistakenly called self-calibration'. Practically:

Operation	What it is
Taring the amplifier	Adjustment - removes an offset from the data, not physical load
ADC internal calibration	Adjustment of the converter, not calibration in newtons
Calibration	A relation from indication to measurand, with uncertainty on both sides

Uncertainty budgets

Type A components are evaluated statistically (repeat loadings); Type B by any other means (mass-standard uncertainty, resolution, hysteresis, creep, temperature). Combine with the law of propagation, **including covariance where components share a source**. Report as $y \pm U$ with the coverage factor k stated, U to at most two significant digits.

Percent of full scale - the trap that drives cell selection

Load cells specify nonlinearity, hysteresis and repeatability as a fraction of **full scale**. Measuring a small force on a large cell therefore gives uncertainty that can exceed the quantity you want.

Worked, for the whole aircraft at the unverified 50 g estimate:

$$\begin{aligned}\text{hover force} &= 0.050 \times 9.80665 = 0.4903 \text{ N} \\ \text{per rotor} &= 0.1226 \text{ N} \\ \text{objective } U &\leq 0.025 \times 0.4903 = 0.01226 \text{ N}\end{aligned}$$

Cell	Full scale	Screen U (3 terms, k=2)	vs objective
5 kg	49.03 N	0.02942 N	2.4x over - not demonstrated compliant
1 kg, if 0.03 %FS class	9.81 N	0.00588 N	0.48x - about 2.1x margin

Derived requirement: **a 1 kg cell must be better than 0.0625 %FS per term** to meet the objective. No accuracy table was verified for the purchased 1 kg part, so that margin is *conditional*. This is a datasheet screen, not measured uncertainty, and neither cell is an accepted calibrated instrument.

Ground effect - why bench clearance is not obvious

For a **whole vehicle**, measured thrust inflation reaches about 1.4x near $z/R = 1$ and does not settle until roughly $z/R = 5$ to 6 - worse than the single-rotor case because downwash from opposing rotors meets under the body (the 'fountain effect'). Partial ground effect produces a pure moment disturbance, which on a bench appears as an apparent pitch or roll bias rather than a thrust error.

Clearance is therefore specified in rotor diameters, in every direction including the ceiling, and the honest move is to *prove* its absence with a height sweep rather than assume it.

Defence question. Your bench reads a stable 0.49 N. Name four reasons that number might not be the aircraft's thrust.

Part 9 - CAD discipline

The pattern is the same for every part:

register -> generator -> analytic oracle -> STEP round-trip -> reviewed contract

with the generator failing closed on any pending input.

Why the oracle is not optional

The oracle is a volume computed by hand, independently of the CAD kernel. It exists because a rendered view cannot show you a dimensional error. In the V995 fixture work it caught two defects that looked perfectly fine on screen:

Defect	How it was caught
A cell hole placed 1.5 mm off the edge of the plate	Measured volume exceeded the analytic volume - material that should have been removed was present
Relief pocket tangent to two bolt holes	Analytic volume subtracted the same material twice: 35.54 mm ³ , exactly the two half-discs

Neither is visible in a screenshot. This is the whole argument for the rule: **a picture is not acceptance; the numbers are.**

Fail-closed, and why pending is not zero

The generator emits a part only when every register row it needs is populated. A pending row is a *missing measurement*, not a zero and not a default. Today the cradle and base plate are skipped, and the generator names exactly which rows are missing.

A test asserts that the **committed** register still cannot build those parts, so a plausible-looking value cannot quietly become a design input. Synthetic dimensions used to exercise the builders live only in the test file.

Design intent that is already encoded

Feature	Why
Cradle is a U-channel, not a gravity pocket	The reaction can reverse; the fixture must transmit uplift as well as downward load
Base plate carries a relief pocket	Keeps the cell's sensing section free to deform; without it the plate is a parallel force path and the
Pocket leaves fastener land	Bolt bearing surface, and it prevents pocket/hole overlap

Defence question. Why does the generator refuse to build the cradle, and why is that better than building it with estimated dimensions?

Part 10 - Literature position

What is established - never claim these

Result	Anchor
Hover and position control after one rotor loss, yaw sac	Friedl/Lanzon/Longhi 2011-14; Mueller and D'Andrea 2014
The relaxed-hover primary-axis mechanism itself	Mueller and D'Andrea - and it is patented (US 9,856,016 B2)
Two-opposite loss, flight-validated above 8 m/s	Sun, Wang, Chu and de Visser, T-RO 2021
All four failure classes in outdoor flight, one non-switching controller	Ke, Contreras-Quan, T-RO 2023
Recovery from arbitrary tumbling after rotor loss	Sun, Baert et al., ICRA 2020 - Monte Carlo plus real flight
Detection-delay distribution, 26 real propeller ejections	Strack van Schijndel et al. 2021 - 95% bounds [28, 132] ms
Fault-tolerant flight on a 30.6 g quadrotor	Cintas and Ozyer 2026 - but PARTIAL degradation, not complete loss

What remains open

The literature is overwhelmingly about **one action**: here is a controller, it stabilises the damaged vehicle. Recovery as a **choice among qualitatively different actions, indexed by post-failure state**, is close to unoccupied. Specifically open:

Gap	Nearest prior work
Action selection over $(h, v_z, \text{tumble rate, detection delay})$	Mo et al. 2024 - two actions on one axis (damage severity) only
The recoverable set as a function of detection delay	Everyone reports achieved latency; nobody publishes the set as a function of it
The abort boundary where the vertical budget runs out	Recovery is shown at generous altitude; the boundary is uncharacterised

Who contests the gap

Siotia et al. (Scientific Reports, 2026) assert in print that the field lacks an integrated framework transitioning between active stabilisation and passive descent using altitude-aware logic. You must cite them and state your difference: theirs is a deployment *threshold* validated in hardware-in-the-loop; yours is a survivability estimate per failure class over a state grid that includes detection delay, with preregistered criteria and exact interval arithmetic.

Defence question. State your contribution in one sentence that survives a reviewer finding one more paper.

Part 11 - Corrections history

This is where real understanding lives. Each item below was a genuine error, found and fixed in the open. Being able to narrate them is what distinguishes someone who owns the work from someone who inherited it.

Correction	What went wrong	The lesson
Parachute terminal speed 2.5 -> 1.8 mm/s	Terminal speed exceeded the impact limit it was judged against	Agree against a device that lands harder than the criterion is implied
Parachute inflation transient added	Deployment was modelled as a delay then instant	Deploy 2.8 times to insist against published data
Allocator single-pass -> cascaded	Pseudo-inverse plus clip distorted torque axis and	Admitted that only appears when the system is stressed - e
partial_authority multiplier -> cap	Coded as $f \neq 0.6$ when it is a per-rotor cap; hover	Airspins below effectiveness loss are different faults
Spin-scaling claim withdrawn	Claimed spin rises as inertia falls; at equilibrium Izz	Workable algebra; intuition about scaling is often wrong
Detection-delay gap claim withdrawn	Claimed no distribution existed; it did, with 95% b	Verify a negative claim as hard as a positive one
Unpaired -> paired comparison	Seeds included the action index, so arms drew different	The fix reverses decision, not repair - the old design was valid
Policy -> design comparison	Actions varied installed hardware; no aircraft choos	Check that every configured option is executable by one sy

Notice the pattern: **most were caught by a check, not by inspection.** The volume oracle, the analytic anchors, the criterion-coherence test, the pinned prediction table. Reading code rarely finds these; running an independent calculation against it does.

Defence question. Pick the correction you find most embarrassing and explain why publishing it strengthens rather than weakens the work.

Part 12 - Defending the work

The four-step answer

When challenged on any number, answer in the order the repository is built:

Step	Question	Where to look
1	What kind of number is it?	Provenance label - requirement, calculation, assumption, measurement
2	Where does it come from?	The file, and the derivation if one exists
3	What if it is wrong?	Sensitivity, and what decisions it blocks
4	How would you close it?	The specific measurement or analysis, and its open question

Challenges to expect, and honest answers

Challenge	Answer
'Your landing criterion is arbitrary.'	It is assumed, and labelled so. Injury data is non-binding at this mass by ~27x. Landing-gear certification is not required.
'You only ran simulations.'	Correct, and no quantity in the repository claims status 'measured'. The bench and drop gates are not required.
'The mechanism result is a null finding.'	It is stronger than null: 69 of 76 discordant pairs favour reallocation, and the mechanism is judged to be a good fit.
'Clopper-Pearson is too conservative.'	Agreed, and cited. The direction is favourable here, and the grid lands near $p=0$ and $p=1$ where all the data is.
'Isn't rotor-out control already solved?'	Yes - and the repository says so, including the patent. The open question is action selection over time.
'Your allocator is suboptimal.'	Yes. It is the redistributed pseudo-inverse, named and cited, justified on implementation cost. Active control is not required.
'How do you know your simulation is right?'	Analytic anchors: hover holds, one-out gives $-g/4$ vertical acceleration, two-out gives $-g/2$, with distributional bounds.

The strongest move available to you

Name the weakness before your examiner does. This project keeps an audit of which numbers are derived, which are asserted and which are bare; an open-question ledger; and a corrections history. Using them proactively converts what looks like a gap into evidence of engineering judgement. **An examiner who finds an unmarked assumption has caught you out; an examiner who finds a marked one has found your system working.**

What you cannot say

Do not say	Because
'The gripper/vehicle is validated'	No physical measurement has been accepted
'90% reliability'	The bounds are conditional sampling bounds under one model
'We proved the mechanism does not help'	Absence of evidence is not evidence of absence; state the interval
'This satisfies 14 CFR 27.725'	That is a different vehicle class; it is context, not compliance
'First to do X'	The literature search is bounded; claim what you searched, not what exists

Every number quoted in this document was read from the repository at compile time. Where a value is assumed rather than measured, that is stated - because that distinction is the entire point of the project.